

THE EFFECT OF MINIMUM WAGES ON LOW-WAGE JOBS*

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We estimate the effect of minimum wages on low-wage jobs using 138 prominent state-level minimum wage changes between 1979 and 2016 in the United States using a difference-in-differences approach. We first estimate the effect of the minimum wage increase on employment changes by wage bins throughout the hourly wage distribution. We then focus on the bottom part of the wage distribution and compare the number of excess jobs paying at or slightly above the new minimum wage to the missing jobs paying below it to infer the employment effect. We find that the overall number of low-wage jobs remained essentially unchanged over the five years following the increase. At the same time, the direct effect of the minimum wage on average earnings was amplified by modest wage spillovers at the bottom of the wage distribution. Our estimates by detailed demographic groups show that the lack of job loss is not explained by labor-labor substitution at the bottom of the wage distribution. We also find no evidence of disemployment when we consider higher levels of minimum wages. However, we do find some evidence of reduced employment in tradeable sectors. We also show how decomposing the overall employment effect by wage bins allows a transparent way of assessing the plausibility of estimates. *JEL* Codes: J23, J38, J88.

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I. INTRODUCTION

Minimum wage policies have featured prominently in recent policy debates in the United States at the federal, state, and local levels. California, Illinois, Massachusetts, New Jersey, and New York have passed legislation to eventually increase minimum wages to \$15/hour, and at least five other states are on paths to raise their minimum wages to \$12 or more. Over a dozen cities have instituted city-wide minimum wages during the past three years, typically by substantial amounts above state and federal standards. Underlying much of the policy debate is the central question: what is the overall effect of minimum wages on low-wage jobs?

Even though nearly three decades have passed since the advent of “new minimum wage research” (see, e.g., [Card and Krueger 1995](#); [Neumark and Wascher 2008](#)), there is surprisingly little research on the effect of the policy on overall employment. This shortcoming is particularly acute given the importance policy makers place on understanding overall responses. For example, in its attempt to arrive at such an estimate, the 2014 Congressional Budget Office (CBO) report noted the paucity of relevant research and then used estimates for teen minimum wage elasticities to extrapolate the total impact on low-wage jobs.

In this article we use a difference-in-differences design to estimate the impact of minimum wage increases on the entire frequency distribution of wages and subsequently focus on changes at the bottom of the distribution to estimate the impact on employment and wages of affected workers. Our approach relies on the idea that the overall employment and wage effects of the policy can be inferred from the localized employment changes around the minimum wage. An increase in the minimum wage will directly affect jobs that were previously paying below the new minimum wage. The jobs shifted into compliance create a “bunching” and show up as “excess jobs” at and slightly above the minimum. The effect of the minimum wage on the wage distribution fades out and becomes negligible beyond a certain point. Therefore, the overall employment and wage effects of the policy can be inferred from the localized employment changes around the minimum wage. For instance, we can assess the changes in employment from the difference between the number of excess jobs at and slightly above the minimum wage and the number of missing jobs below the minimum.

To identify the effect of the minimum wage on the frequency distribution of wages, we implement an event study analysis that exploits 138 prominent state-level minimum wage increases between 1979 and 2016. We estimate employment changes in each dollar wage bin relative to the new minimum wage for three years prior to and five years after an event. Our empirical approach therefore disaggregates the total employment effect of the policy into constituent wage bins, and we use these bin-by-bin estimates locally around the minimum wage to assess the effect of the policy.

There are several advantages of our disaggregated approach relative to the more standard approach that estimates the disemployment effect using aggregate employment or wage changes (e.g., [Meer and West 2016](#)). First, we focus on employment changes locally around wage levels where minimum wages are likely to play a role. When only a small fraction of the aggregate workforce is affected by the minimum wage, such a localized approach is crucial for uncovering meaningful “first-stage” wage effects of the minimum wage—something that is not possible with the standard approach except for subgroups like teens. Second, by decomposing the aggregate employment impact by wage bins, we are able to assess employment changes in the upper tail of the wage distribution. This can provide an additional falsification test, since large changes in the upper part of the wage distribution are unlikely to reflect a causal effect of the minimum wage. Third, our localized focus on jobs around the minimum wage gains precision by filtering out random shocks to jobs in the upper part of the wage distribution.

We use hourly wage data from the 1979–2016 Current Population Survey to estimate the effect of the minimum wage by wage bins. We find that an average minimum wage hike led to a large and significant decrease in the number of jobs below the new minimum wage in the five years after implementation. At the same time, there was clear evidence for the emergence of excess jobs at or slightly above the minimum wage. However, as expected, we find no indication of any employment changes in the upper part of the wage distribution—providing further validation to the empirical design. We estimate that the number of excess jobs closely matched the number of missing jobs: the employment for affected workers rose by a statistically insignificant 2.8% (std. err. 2.9%). Our estimates also allow us to calculate the impact of the policy on the average wages of affected workers, which rose by around 6.8% (std. err. 1.0%). The significant increase in average wages of

affected workers implies an employment elasticity with respect to own wage (or the labor demand elasticity in a competitive model) of 0.41 (std. err. 0.43), which rules out elasticities more negative than -0.45 at the 95% confidence level.

An additional advantage of estimating the effect of the minimum wage on the frequency distribution of wages is that we can assess the extent to which the direct wage effects of the minimum wage are amplified by wage spillovers. We find that spillovers extended up to \$3 above the minimum wage and represent around 40% of the overall wage increase from minimum wage changes. Interestingly, we also find that the benefits of wage spillovers were not equally shared: workers who had a job before the minimum wage increase (incumbents) experienced significant wage spillovers, but we do not find any evidence of such spillovers for new entrants. This asymmetry suggests that spillovers may reflect relative pay concerns within the firm (Dube, Giuliano, and Leonard 2018) and the value of outside options or reservation wages of nonemployed workers is unlikely to play a key role in generating wage spillovers (e.g., Flinn 2006).

Our estimates are highly robust to a wide variety of approaches to controlling for time-varying heterogeneity that has sometimes produced conflicting results in the existing literature (e.g., Allegretto et al. 2017; Neumark and Wascher 1992). Moreover, the shifts in the missing and excess jobs are strongly related to the timing of minimum wage change—providing further support that we are identifying the causal effect of the policy. Both missing jobs below the new minimum and excess jobs above were close to zero prior to the minimum wage increase, which suggests that the treatment and the control states were following a parallel trend. The drop in jobs below the minimum wage is immediate, as is the emergence of excess jobs at or slightly above the minimum. Over the five-year post-treatment period, the magnitude of the missing jobs below the new minimum wage decreases only slightly, underscoring the durability of the minimum wage changes studied here.

To go beyond our overall assessment of the 138 case studies used for identification, we also produce event-by-event estimates of the minimum wage changes. Although we find substantial heterogeneity in the bite of the events, the distribution of employment effects are consistent with a sharp null of zero effect everywhere. For example, our event-by-event analysis finds that the estimated missing jobs rose substantially in magnitude with the